Principles of Statistics

Part II

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Principles of Statistics

0. INTRODUCTION

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and $X : \Omega \to \mathbb{R}$ a random variable with distribution function

$$F(t) = \mathbb{P}\big(\{\omega : X(\omega) \leq t\}\big) = \begin{cases} \displaystyle\int_{-\infty}^{t} f(x)\,dx, & \text{if } X \text{ is continuous with pdf } f, \\ \displaystyle\sum_{x \leq t} f(x), & \text{if } X \text{ is discrete with pmf } f. \end{cases}$$

We denote $P = \mathbb{P} \circ X^{-1}$, the law or "distribution" of X.
We observe $X_1, \ldots, X_n$, which are iid copies of X, where n is the sample size.

Def 1. A statistical model is a collection $\{f_\theta : \theta \in \Theta\}$ of pdfs or pmfs, or $\{P_\theta : \theta \in \Theta\}$ of laws, indexed by a parameter space Θ.

Ex.

- $N(\theta, 1)$, $\Theta = \mathbb{R}$.
- $N(\mu, \sigma^2)$, $\theta = (\mu, \sigma^2)$, $\Theta = \mathbb{R} \times (0, \infty)$.
- $\text{Poi}(\theta)$, $\Theta = [0, \infty)$.

Def 2. We say the model is correctly specified when $\exists\, \theta_0 \in \Theta$ such that $P = P_{\theta_0}$.
We will refer to the "true parameter" as θ_0.

Goals of Statistics

1. Estimation: We define an estimator
$$\widehat{\theta}_n = \widehat{\theta}_n(X_1, \ldots, X_n),$$
which we hope is close to θ_0 when $X_i \overset{\text{iid}}{\sim} P_{\theta_0}$, for all $\theta_0 \in \Theta$, when n is large.
2. Testing: We have a null hypothesis
$$H_0 : \theta \in \Theta_0 \subseteq \Theta,$$
and an alternative hypothesis
$$H_1 : \theta \in \Theta_1 \subseteq \Theta.$$
We define a test,
$$\psi_n := \psi_n(X_1, \ldots, X_n),$$
such that $\psi_n = 0$ if H_0 is true, and $\psi_n = 1$ if H_1 is true, w.h.p.

3. Uncertainty Quantification

- Define a confidence set $C_n(X_1, \ldots, X_n; \alpha)$, such that

$$P_\theta(\theta \in C_n) \overset{(\geq)}{=} 1 - \alpha \qquad \forall\, \theta \in \Theta,$$

  where α is the significance level.
- Define a Bayesian credible set, a region of high posterior probability.

I. LIKELIHOOD PRINCIPLE

Def 3. The likelihood function

$$L_n(\theta) = \prod_{i=1}^{n} f_\theta(X_i).$$

Note: This is a random function, as it depends implicitly on $X_1, \ldots, X_n$.

The log-likelihood function

$$\ell_n(\theta) = \sum_{i=1}^{n} \log f_\theta(X_i),$$

we write

$$\overline{\ell}_n(\theta) = \frac{1}{n}\ell_n(\theta)$$

for a normalised log-likelihood.

Def 4. A maximum likelihood estimator $\widehat{\theta}_{\text{MLE}}$ is any estimator with

$$L_n\big(\widehat{\theta}_{\text{MLE}}\big) = \max_{\theta \in \Theta} L_n(\theta).$$

Ex.

$$X_i \overset{\text{iid}}{\sim} \text{Poi}(\theta).$$

$$L_n(\theta) = \prod_{i=1}^{n} e^{-\theta} \frac{\theta^{X_i}}{X_i!}.$$

$$\ell_n(\theta) = -n\theta + (\log \theta) \sum X_i + \text{const.}$$

$$\ell_n'(\theta) = -n + \frac{\sum X_i}{\theta} \overset{\text{FOC}}{=} 0$$

$$\implies \quad \text{suggests } \widehat{\theta}_{\text{MLE}} = \frac{\sum_{i=1}^{n} X_i}{n} := \overline{X}_n \qquad \text{“sample mean”}.$$

SOC:

$$\ell_n''(\theta) = -\frac{1}{\theta^2} \sum X_i < 0 \qquad \text{if } \sum X_i > 0.$$

If $\sum X_i = 0$,

$$\ell_n(\theta) = -n\theta \quad \implies \quad \widehat{\theta}_{\text{MLE}} = 0.$$

In every case

$$\widehat{\theta}_{\text{MLE}} = \overline{X}_n.$$

Ex.

$$X_i \overset{\text{iid}}{\sim} N(\mu, \sigma^2), \qquad \theta = (\mu, \sigma^2).$$

$$\ell_n(\theta) = -\frac{n}{2}\log \pi - \frac{n}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(X_i - \mu)^2.$$

FOC:

$$\frac{\partial \ell_n}{\partial \mu} = -\frac{1}{\sigma^2}\sum_{i=1}^{n}(X_i - \mu) = 0,$$

$$\frac{\partial \ell_n}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2}\sum_{i=1}^{n}(X_i - \mu)^2 = 0.$$

We can check that the Hessian of ℓ_n is negative definite $\forall\, \theta \in \mathbb{R} \times (0, \infty)$.
Hence,

$$\widehat{\theta}_{\text{MLE}} = \begin{pmatrix} \overline{X}_n \\ \underbrace{\frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X}_n)^2}_{\text{“Sample variance”}} \end{pmatrix}.$$

Non-examinable

The likelihood principle poses that if two observations

$$X = (X_1, \ldots, X_n) \qquad \text{and} \qquad X^* = (X_1^*, \ldots, X_n^*)$$

give rise to likelihoods L_n, L_n^* respectively, such that

$$L_n(\theta) = L_n^*(\theta) \cdot M \qquad \forall\, \theta \in \Theta,$$

and some constant M, then they should produce the same “inference” on the true parameter.

- The MLE, Bayes estimators, credible sets satisfy the likelihood principle.
- Confidence sets, tests generally do not.

Birnbaum (1962)

$$\text{Conditionality} + \text{Sufficiency} \quad \Longrightarrow \quad \text{Likelihood principle.}$$

end of lecture 1